GEOMETRY PROPOSES, JUDGEMENT DISPOSES: ENGINEERING ONLINE CLAIM AGGREGATION FOR LIVE DELIBERATION

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ABSTRACT

Online deliberation platforms let participants answer an open question in free text. Left raw, fifty people asking for the same thing in fifty wordings look like fifty minority positions, so the platform must decide, live and one statement at a time, whether a new statement *is* an existing proposal, merely *relates* to one, or is new. We describe the aggregation engine of a deployed platform and the design law it converged on empirically: *cosine similarity over text embeddings is a candidate generator, never a decision rule*. Every placement that changes what the public sees is a discrete, fail-closed, cached LLM judgement over texts that geometry proposed, and an opt-in *claim registry* reframes deduplication as classification against a shared, growing codebook of canonical claims. On the 875 hard preference triplets of Blair et al. (2026), the production embedding scores 0.5% triplet accuracy and 99–100% of stance-flipped distractors clear every merge threshold, while the judged registry reaches 95.0% (96.7–98.0% on a subsample under a newer model), against 80–81% for the best repaired geometry. On a 100-statement live benchmark streamed through the real trigger path, the synthesis layer recovers 150/150 ground-truth pairs with zero false merges across three arrival orders. We then examine the opposite shortcut, handing the corpus to an LLM: on that clean corpus a bare model is just as exact at synthesis and better at theming, which locates the architecture's value in scale, streaming and cost rather than per-decision accuracy; but list-scale classification degrades from 95% to 36–56%, canonicalisation is unstable across model upgrades, repeated judgement of a static set converges to spurious merges, and outages masquerade as verdicts unless the system is built to tell them apart. We also report the failure modes that shaped the architecture, a real-data replay with blind audit, cost controls, and how our decision layer composes with Blair et al.'s preference geometry.

1 INTRODUCTION

A growing family of systems for collective decision-making asks participants to express views as free text rather than votes on a fixed slate: Polis (Small et al., 2021), Remesh, the Habermas Machine (Tessler et al., 2024), and generative social choice (Fish et al., 2026; Boehmer et al., 2025). All need the same primitive: given many free-form statements, decide which ones say the same thing. Two shortcuts present themselves: embed each statement and merge whatever lies within a cosine threshold, or hand the corpus to a large language model and ask for groups. We deployed a live deliberation platform that had to answer this question in production for tens of thousands of statements in English and Hebrew, and this paper reports what we learned: both shortcuts fail, for different reasons, and the working architecture is neither.

The geometric shortcut fails in a way Blair et al. (2026) make formal: off-the-shelf embeddings encode a preference-relevant signal entangled with semantic nuisance (topic, wording, style), and on natural data the two correlate, so cosine appears to track stance until the correlation is broken. Their remedy is to repair the geometry; ours, arrived at independently from production incidents, is to demote it. The two are directly comparable because we benchmarked our decision layer on their released hard-triplet set (§5.1).

The LLM shortcut fails less visibly, and we are careful about what fails. An LLM is an excellent judge of a *pair* or a short candidate list, and on a clean corpus that fits in one prompt it is an exact clustering algorithm (§5.4). What does not survive production is the operating regime: accuracy collapses as the candidate list grows (§3.2), free-form canonicalisation drifts across model generations (§5.1), repeated "find groups" judgement of a static set converges to spurious merges (§4.6), per-arrival cost is quadratic in corpus size, and failures are silent unless the system separates outages from verdicts.

Contributions. (1) A production architecture for *online* claim aggregation in which geometry only proposes candidates and every placement is a discrete, fail-closed, verdict-cached LLM judgement (§4). (2) The *claim registry*: deduplication as classification against a growing per-question codebook, with consolidation, a mutation protocol, self-audit counters and typed *opposes* edges no metric can represent (§4.5). (3) Evaluation on Blair et al.'s 875 hard triplets, a live end-to-end benchmark in two languages, a production replay with blind audit, and LLM-only baselines (§5). (4) A catalogue of failure modes of embedding-threshold clustering and the structural answer to each (App. F). (5) Why our decision layer and Blair et al.'s geometry are complementary, and the hybrid this suggests (§6).

2 Problem Setting

Participants answer an open question (e.g. *"What should our city do to improve residents' quality of life?"*) with short proposals and evaluate each other's proposals on a continuous scale. The engine's job is threefold and continuous: **synthesis**, detect that two statements are the *same proposal* in different words and replace them with one jointly-worded statement that pools their evaluations; **theming**, group distinct-but-related proposals under topical headings without pretending they agree; **standing**, leave everything else alone, visibly, from statement #1. Formally, statements $x_1, x_2, \ldots$ arrive in an arbitrary order, the state S_t is a set of clusters over $\{x_1..x_{t-1}\}$, and the unit operation is *given x_t and S_t, decide where x_t belongs*: an online placement problem, not a one-shot clustering problem, under four constraints. It runs *live*, in seconds, with no batch step a user waits for; arrival order is not under our control and must not change what people are agreeing to; corpora are per-question, 10^2–10^4 statements, in several languages; and LLM spend must be bounded and auditable.

Two distinctions do most of the conceptual work. *Same topic $\neq$ same proposal*: "add speed bumps near schools" and "lower the speed limit near schools" share a goal but are different interventions; merging them misrepresents both authors. *Same words $\neq$ same stance*: "ban X" and "do not ban X" are lexically near-identical and preferentially opposite. The errors are asymmetric: a wrong merge silently files a statement under a proposal its author would reject and is permanent for every reader; a missed merge leaves a visible duplicate a later sweep can repair. This asymmetry is the source of every tie-breaking bias in the system.

3 Why Neither Geometry Nor a Bare LLM Suffices

3.1 Geometry: a topicality signal wearing a preference costume

Blair et al. decompose the cosine margin between an anchor a, a preference match p and a semantic distractor n into a stance component and a nuisance component, $s(a,p) - s(a,n) = \Delta_S + \Delta_T$, weighted equally; on hard triplets (a distractor that keeps the wording and flips the stance) the nuisance dominates and base encoders rank the distractor above the match 52–94% of the time. Our production measurements agree from the other direction (Table 1): with `text-embedding-3-small`, **100% of stance-flipped distractors clear the pipeline's topical gate (0.60) and 99.1% clear its strictest merge gate (0.85)**, and the distractor median cosine (0.969) sits *above* the match median (0.871). A threshold that admits every counterfeit is not a classifier. Three production facts compound this: a shared question prefix lifts the whole cosine floor (80% of English and 100% of Hebrew *cross-topic* pairs on our live corpus sit above 0.60); thresholds are properties of a geometry, not a pipeline (bands calibrated for one model re-opened a catastrophic failure when the Hebrew model was upgraded, App. F); and theme membership is not in the geometry at all (the best cosine cut over ground-truth synthesis centroids reaches topic F1 0.48; a judge took it to 0.78). Yet nearest-neighbour search over the same embeddings reliably *surfaces* the true partner: the paraphrase twin is the top neighbour for 100/100 English statements, and at a 0.45 retrieval floor no true match was lost in any language tested

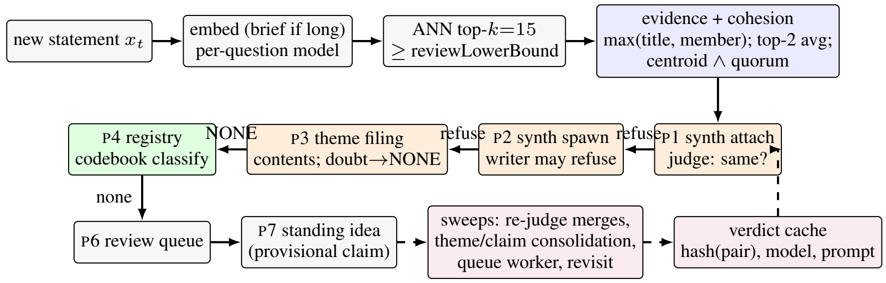

Figure 1: The placement cascade. Blue: geometric gates (candidacy only); orange: LLM judgements, the only steps that change what the public sees; green: the opt-in registry; purple: the convergence layer. A statement takes the first pass whose gate *and* judgement both admit it.

(§5.1). Geometry knows what is *relevant* even when it cannot tell *which side*; that asymmetry is the whole design.

3.2 A BARE LLM: AN EXCELLENT JUDGE, A POOR CLUSTERING ALGORITHM

The natural reaction is to skip geometry and let an LLM decide everything. On a corpus small enough to fit in one prompt this works, and §5.4 says so with numbers; what follows is why it does not survive the production regime, each item on a different axis. *Scale and cost.* Pairwise judgement of n statements is $\Theta(n^2)$ calls; a sweep that lacked a spend bound on refusals cost ≈3,100 completions per day on proposals nobody was touching (§5.4). Placing each arrival against the whole corpus costs $\Theta(n)$ tokens per arrival, $\Theta(n^2)$ overall, bounded by the context window; one-shot batch clustering has no online form, since a rebuild on every arrival can change what earlier participants agreed to. *List-scale degradation.* The judge that scores 95.0% against a single candidate claim scores 36–56% under strict scoring when it must pick among a ∼94-claim codebook (Table 1), consistent with position-bias and long-context findings (Liu et al., 2024; Wang et al., 2023); a judge over a short ranked list and one over a long list are not the same instrument. *Unstable canonicalisation.* Asking a model to write a “simple meaning” sentence per statement and comparing the sentences recreates the problem one level up: two calls canonicalise one idea two ways. Classification into a fixed set converges; free-form generation does not. Even with the codebook fixed, replacing raw text by a generated canonical cost 24.9 points (95.0% → 70.1%), and a routine model upgrade later degraded the enriched-canonical configuration by 12 points while leaving raw-text classification intact (§5.1). *Non-determinism and looping.* At temperature 0 the models are not deterministic (Atil et al., 2024); a sweep asked to *find* merge groups in a settled, correct set of ten headings proposed a wrong merge in one sample of two, and looped ∼144 times a day, as production does, a rare spurious merge becomes a near-certainty (§4.6). *Prompt sensitivity.* With the architecture fixed, a plausible hand-written judge prompt scored 36% triplet accuracy with 43% false merges where the production prompt scored 79% and 7% (App. B); the difference is two defensive clauses and a confidence floor, invisible without an adversarial test set (Sclar et al., 2024). *Silent failure.* An exhausted rate limit that “fails closed” to *none* is indistinguishable from a burst of honest “this is new” verdicts (260 of the first 456 classifications in one run were silent corruption), and reasoning models can exhaust a completion cap on hidden tokens and return empty JSON deterministically; outages must be tracked separately from verdicts, fallbacks never cached, every dropped action given a custodian. And a batch partition is a fait accompli, whereas participants are entitled to a stated reason and a typed relation for every placement.

The conclusion is not “don’t use an LLM” but a division of labour: geometry proposes a short, ranked candidate set; the LLM disposes with a discrete verdict over texts, under an asymmetric doubt bias, fail-closed, cached, budgeted on cache misses, and logged with its reason.

4 THE SYSTEM

4.1 REPRESENTATION AND RETRIEVAL

Statements are embedded with OpenAI `text-embedding-3-small` at 1536 dimensions (OpenAI, 2024); a stronger model (`-3-large`, truncated natively to the same width) is pinned *per*

Algorithm 1 Online placement of one statement (abridged; every pass is fail-closed on judge error).

$e \leftarrow \textsc{Embed}(x_t)$; $C \leftarrow \text{ANN}(e, k{=}15, \geq \text{reviewLowerBound})$
if $C = \emptyset$ **then return** $\textsc{Registry}(x_t)$ **or** provisional singleton claim
end if
$E \leftarrow$ best evidence per cluster: $\max$(title, member), promoted to top-2 member average
for all synth s with $E[s] \geq$ attach, sorted desc **do** ▷ P1
  if member evidence $<$ attach **or** $\neg\textsc{Cohesion}(s, e)$ **then continue**
  if $\textsc{JudgeCached}(x_t, s) = \textsc{same}$ **then** attach; re-home theme; **return**
end for
for all plain $y \in C$ in synth band, not in a synth, first 3 **do** ▷ P2
  $r \leftarrow \textsc{Writer}(x_t, y)$; **if** spawned **then** nest under theme; **return**
  if debounced **or** error **then** enqueue retry; **return** ▷ never drop
end for ▷ refusal: try next candidate
if not in a theme: $\theta \leftarrow \textsc{ThemeJudge}(x_t$, themes with contents); **if** $\theta \neq \textsc{none}$ attach; **return** ▷ P3
if registry on: $r \leftarrow \textsc{Classify}(x_t$, codebook ordered by cosine); record OPPOSES edge; **if** EXPRESSES$\wedge$conf$\geq$0.6 attach; **return** ▷ P4
queue $(x_t, \text{top}(C))$ for review ▷ P6; resolved automatically if later placed

question, since vectors from different models occupy different spaces and a cosine between them is meaningless. Hebrew forces the pin: under the small model the paraphrase twin is the nearest neighbour for only 56/100 Hebrew statements (100/100 English), 88–89/100 under the large one. Long statements are first distilled into a 5–15-word *brief*, which is what gets embedded; boilerplate otherwise makes everything look $\sim$0.75-similar. Candidates come from approximate nearest-neighbour search over the question's partition, top $k = 15$ (raised from 10 because compressed Hebrew geometries buried true partners below rank 10). Cosine is partitioned into calibrated bands (App. A) with the invariant review $<$ cluster $\leq$ synthLower $\leq$ attach; crossing a band makes a statement a *candidate* for an action, never triggers it.

4.2 Evidence and cohesion gates

Evidence for a candidate cluster is deliberately not "similarity to the cluster title". Merged titles abstract away from their members, so title-only evidence spawns duplicate clusters; but title-*or*-any-member evidence lets a cluster *snowball*, absorbing a chain of pairwise neighbours until it spans several ideas (observed in production: one "safe settlement" synthesis absorbed belonging, beauty, growth and community, one member riding in at cosine 0.43 to the rest). We therefore compute candidacy as $\max(\text{title cosine}, \text{best member cosine})$ for discovery, then fetch *all* member embeddings and score the cluster by the *average of its top-2 member cosines*, so two independent members must agree, with member-only evidence tracked separately so the strictest gate can demand agreement with the cluster's *contents*. Before any merge candidate reaches a judge, the receiving cluster must also pass a cohesion gate combining, by conjunction, cosine to the member centroid and a quorum of members above a floor. The centroid floor (0.815 at defaults) comes from a measured separation over 4,900 candidates (genuine attaches: median 0.898, p_{10} 0.862; false same-topic: median 0.723, p_{99} 0.831); as an OR with lower floors the gate never once refused an attach in a 100-statement run, and the conjunction is what makes it a brake.

4.3 The placement cascade

Algorithm 1 gives the decision tree; one function serves all five entry points (live creation, threshold crossings, the retry queue, admin re-runs), which is what makes the behaviour analysable. Three ordering rules were each learned from a measured failure. *Most-specific-first*: synthesis passes run before theming, because a theme that already holds your twin will always look cohesive to you (your twin is inside its centroid); with theming first, 17 of 23 topical attachments in one run were justified by the statement's own paraphrase twin, and 2 of 50 pairs ever merged. *Topic membership is non-terminal*: a statement owned by a theme has been given "a theme, not a meaning" and remains eligible for synthesis when its twin arrives; synthesis membership *is* terminal, and an authoritative reverse-membership query enforces exactly-one-owner idempotence under queue replays. *Judged attach, everywhere*: P1 was the last placement that acted on geometry alone, and under Hebrew with the large model distinct transport ideas attached to one synthesis at cosine 0.899–0.944, above every gate; the equivalence judge now gates P1 too, verdict-cached and fail-closed, because an unmerged duplicate self-heals via the sweep and a wrong attach is permanent.

4.4 THE JUDGES

Every consequential action is an LLM judgement over the actual texts with three recurring safety properties. *Asymmetric doubt*: each prompt carries an explicit tie-break toward the cheaper error (the equivalence judge: "when in doubt between same and related, prefer related"; the writer: "when in doubt, refuse"; the theme judge inverts it: "when unsure, answer NONE: a topic created too eagerly is cheap, a proposal filed under the wrong topic stays there for every reader"). *Fail-closed placement, fail-open custody*: an error on a merge decision defaults to *different*; an error on re-validation of an existing cluster keeps its members; outages are tracked separately from negative verdicts. *Reversible by sweep, not irreversible by stream.*

The **equivalence judge** returns a four-way verdict, SAME — RELATED — DIFFERENT — OPPOSITE (App. B); only SAME merges, and OPPOSITE is a typed relation a distance function cannot represent. The **synthesis writer** produces the joint statement under a rubric whose first stage is a refusal check (directional conflict; *distinct actions within the same theme*), amended twice on measured errors: a *formulation clause* (refuse only when the difference *is* the intervention), which took Hebrew twin recovery from 45/50 to 49/50 without moving English, and a *never-widen-scope rule*, after synthesized texts drifted toward helpful generalisation ("freelancers" → "residents"), which *silently changes what people are agreeing to*: scope inflation went 4/50 → 0/50 and a live fidelity audit measured 98/98 member intents preserved (App. H). Output language is forced by script detection, since titles are themselves embedded and a wrong-language title is a retrieval corruption. The **theme judge** files a statement into an existing theme shown with its *contents*, never by title alone, may answer NONE, and never creates a theme. The two factors only work together: contents under a "prefer an existing topic" bias made misfiling *worse* (28 → 37 on a certified replay), caution without contents refused half of everything, and together permanent misfiles halved live (20.7% → 9.3%). The selection rule, registered before the numbers landed, minimises $3 \cdot$ misfiles + over-NONEs; two variants with higher per-decision *accuracy* were rejected for re-licensing misfiles (App. G), since accuracy is the wrong objective for an asymmetric-cost decision.

4.5 THE CLAIM REGISTRY

The cascade shares a structural blind spot: the judge can only rule on pairs that cosine surfaced, so two statements with the same meaning but distant embeddings are never compared, a *semantic recall gap* no threshold tuning can close. A second problem is cold start: cosine bands are density-calibrated and mean little when a question has five statements. The registry answers both with one move: it reframes deduplication as **classification against a shared, growing codebook of canonical claims**, one per idea already seen in this question, each carrying a 5–15-word canonical wording, a public explanation and a verbatim exemplar. Because claims are short, one fast-model call reads *all* of them, so recall is independent of geometry; because classification is into a fixed set, canonical forms converge instead of drifting; and because meaning-judgement is calibration-free, it works as well at statement #2 as at #2000. Enriched lines matter (bare canonicals cost 25 points, §5.1), as does the stance caution learned from the hard triplets. Cosine is demoted from gatekeeper to *ranker*: it orders the codebook most-plausible-first, mitigating position bias, and filters nothing. Singletons spawn a one-member *provisional* claim, so every idea is a labelled public idea from the moment it exists.

The maintenance layer treats the codebook as an instrument that must not drift. *Consolidation*: every 15 placements, one call over the whole codebook proposes merges and too-broad flags under a conservative rubric ("same-topic-different-action is NOT a merge; no merges is a perfectly good answer"); each claim joins at most one merge per pass (fixed-point iteration), membership transfers transitively, splits are queued for humans, and claims are confirmed only with $\geq$3 members after surviving a pass untouched. *Mutation protocol with a broaden ratchet*: any rewording is itself classified (*reword — broaden — narrow — different*, failing closed to *different*, which forces one batched re-validation over member briefs); because innocent broadenings compose, every claim keeps its *anchor text* and after two unchecked broadenings the new wording is tested against it. *Self-audit*: every consolidation merge is by construction an observed false-"new", so the running merge count is an online estimate of the registry's own recall error, otherwise structurally silent; 5% of classifications are re-run on a stronger model and logged. *Scale path*: above 30 claims classification routes through at most two topic claims, but any NONE triggers a full flat fallback, because *a gate that filters can misfile; a gate with an ungated second look cannot* (hard partitioning without fallback caused 59% of

Table 1: Triplet accuracy on the 875 hard triplets of Blair et al.. Paper rows are the authors' reported cosine accuracies on the same instrument; Freedi rows are end-to-end attach decisions. McNemar registry vs raw cosine: 827 discordant wins to 0.

Condition	Decision mechanism	Acc. [95% CI]
Cosine, raw text (production model; 1.8% with the question prefix)	threshold on `text-embedding-3-small`	0.5% [0.2, 1.2]
distractor ≥ 0.60 / ≥ 0.85 (false attach at P3 / P1)		100.0% / 99.1%
Base ST5-XL / e5-large-v2 / BGE-large / all-mpnet (paper)	cosine	48.3 / 26.7 / 6.3 / 8.2%
DPT-tuned ST5-XL (paper)	cosine on preference-tuned encoder	80.0%
Per-topic rank-20 projection (paper)	vote-trained ideal-point scorer	81.1%
Registry, generated 5–15-word canonicals	LLM classification vs bare codebook	70.1% [66.9, 73.0]
Registry, enriched lines + stance caution	+ explanation + exemplar per claim	88.6% [86.3, 90.5]
Registry, raw-anchor claims (production worker; 96.5% on a stronger model)	LLM classification	**95.0% [93.3, 96.2]**
Embedding gate (≥ 0.45 or ≥ 0.60) then judge	RAG-style	95.0% (0 lost at gate)
Same judge under list-scale competition: full per-dataset codebooks, mean 94 claims per prompt, unconsolidated; strict = own claim		
Bare canonicals	match→any 74.4%, false merge 10.3%	36.0%
Enriched lines + stance caution	match→any 89.8%, false merge 10.3%	53.8%
Two-hop hierarchical + flat fallback ($n = 613$)	match→any 92.2%, false merge 8.2%	55.8%

recall loss in a corpus replay; the fallback moved accuracy $61.3\% \rightarrow 75.3\%$, $p = 7.5 \times 10^{-4}$). The registry is opt-in per question; the judged cascade is the default.

4.6 Convergence and cost control

The live path optimises latency and safety; scheduled functions supply eventual consistency (App. C). A *queue worker* drains deferred work every minute; nothing may be dropped without an audit record or a scheduled second chance. A *re-judge sweep* proposes cross-synthesis merges wherever cross-member top-2 evidence exceeds 0.82, deliberately *below* decision grade because the "one proposal, two wordings" and "two proposals, one topic" bands overlap at 0.82–0.86; each proposal is confirmed by the judge, and as an anti-snowball check the *farthest* cross-member pair must also be judged SAME. *Theme and claim consolidation* judge on *members*, never titles ($17 \rightarrow 10.5$ headings with 6 correct merges vs 3.5 on titles, at equal precision), and each distinct set is judged *once*, keyed on a fingerprint: a settled question otherwise puts the same question to a non-deterministic model ∼144 times a day, and a spurious merge any single call rarely proposes becomes a near-certainty across enough calls.

LLM spend is controlled by a *verdict cache* keyed by an order-independent hash of the pair's texts and versioned by model and prompt (fallback verdicts from failures are never written; a hit costs ∼45 ms against ∼2.3 s for a completion); by *budgets billed on cache misses only* (the sweep's per-question allowance counts judge calls, not successful merges, since refusals cost a call and left the old loop with no spend bound at all, and counts only misses, since billing hits would stall merging on the largest questions); by batching (20 pairs per judge call, one call per cluster re-validation); and by the fail-closed/fail-open policy table of App. A, where `failedClosed` is a first-class marker so a saturated rate limit looks like an outage, not a burst of new ideas.

5 Evaluation

5.1 Hard triplets: a shared instrument with Blair et al.

Blair et al. release 875 hard triplets over 11 datasets from Polis, Remesh and generative-social-choice surveys: for each real participant anchor, a same-stance paraphrase (*match*) and a stance-flipped rewrite preserving the wording (*distractor*). Their metric is a ranking, $s(a,p) > s(a,n)$. Ours is stricter and end-to-end: a triplet counts only if the match *attaches* to the anchor's claim (EXPRESSES at confidence ≥ 0.6) *and* the distractor does not. The harness imports the production functions unmodified; intervals are Wilson 95%; contrasts are exact McNemar on paired triplets.

Three caveats matter. (1) The 24.9-point drop from raw-anchor to generated-canonical localises the weak link in *canonicalisation*, not classification (222 fixed vs 4 regressed, $p \approx 2 \times 10^{-60}$): compression shifts the proposition and the classifier faithfully misjudges against shifted text, which is why production claims carry exemplars. (2) An embedding-gate-then-judge baseline *ties* the registry at 95.0% here, because hard triplets are lexically close by construction; the headline gain is the judge's, and the registry's marginal value is cold start, cross-lingual recall and the maintained codebook. (3) The distractors were generated to defeat embeddings; nothing in their construction targets a model that reads the text, and distractors built against a judge (buried negation, scope shifts, hedges, sarcasm) could score materially worse, so we treat 95% as an upper bound of unknown tightness. The judge also *names* the relation: 95.9% of distractors are labelled OPPOSES, which is what makes counter-edges free; the stronger model prefers NONE (80.6%) and yields worse counter-edge coverage at $\sim$20$\times$ the price.

Robustness across model generations. Re-running the registry conditions on a fixed 150-triplet subsample after a model upgrade (App. D) left raw-anchor classification unchanged (96.7% vs 98.7%, $p = 0.45$; 98.0% after a reasoning-token headroom fix) but regressed the enriched-canonical condition from 92.0% to 79.7% ($p = 2.8 \times 10^{-4}$), because the new model's canonicals dropped stance qualifiers ("illegal, with a life exception" became "permitted when the mother's life is at risk"), after which the classifier *correctly* matched the flipped distractor to a stance-lossy claim. Compressive canonicalisation is inherently lossy; the robust configuration keeps canonicals for *display*, matches on the verbatim exemplar, and regression-tests canonicalisation on every model upgrade.

The recall gap, probed directly. Hard triplets cannot measure the registry's recall claim, so we generated adversarial same-stance rewrites of 150 anchors (metaphorical, colloquial, values-framed, formal; lowest-cosine candidate first, stance verified independently) and Hebrew translations. No English rewrite could be pushed below the 0.60 gate (min cosine 0.738). Six of 125 verified Hebrew pairs fell into the 0.45–0.60 band, where the gated pipeline recovers 0/6 and the ungated registry 6/6 (100% overall). The gap is real but narrow and mostly cross-lingual, and the judge's 2.1% false-attach rate shows the low-cosine recall is not bought with indiscriminate attaching.

5.2 LIVE END-TO-END BENCHMARK

The full pipeline was evaluated on a synthetic-but-adversarial corpus: 100 statements = 10 topics $\times$ 5 idea-groups $\times$ 2 near-paraphrases, streamed one at a time in seeded-shuffled order through the real trigger path in the emulator, with a structurally identical Hebrew translation. Ground truth is 50 syntheses in 10 themes; the metric is $0.6 \cdot F1_{\text{synth}} + 0.4 \cdot F1_{\text{topic}}$, both pairwise over all 4,950 pairs, so under- and over-merging are both punished and coverage folds into recall (the full campaign arc is plotted in App. E).

The first instrumented run scored 0.067: one early topical cluster absorbed all 100 statements (audit: 1 spawn, 98 attaches) and *zero* syntheses formed, while coverage read 100/100; the one number that looks healthy on a dashboard was the one hiding the failure (App. F). The final English configuration reaches **0.884–0.910** across seeds $\{42, 7, 1234\}$, and the property we consider most important is a *separation of variance by layer*: across three arrival orders the synthesis layer recovered **150/150 ground-truth pairs with zero false merges in 450 placements**, exact and order-independent ($P = R =$ ARI $= 1.000$), while the theming layer remains order-sensitive (topic F1 0.711–0.775, 10–17 headings for a true 10). The layer that changes what people are agreeing to is deterministic-in-effect; the layer that only organises the page absorbs the stochasticity. Hebrew reached $\approx$**0.93** (synth $P = 1.000$, 0 false merges, 45/50 pairs, ten themes for a true ten) by one law applied repeatedly: each fix found the next place geometry acted alone and put the judge in front of it; single-run composites cannot resolve a two-pair improvement under $\pm$0.05 filing variance, so we quote ranges. *Cross-generator test:* on a second 10$\times$5$\times$2 corpus authored by an independent vendor's model with harder geometry (cosine-only ceiling F1 0.871), the judged pipeline scored synth F1 **0.980,** $P = 1.000$**, 48/50, 0 false merges**, beating the cosine ceiling by 11 points on a corpus its authors did not write (composite 0.791, dragged by genuinely overlapping domains at topic F1 0.508).

5.3 A REAL PRODUCTION QUESTION, REPLAYED AND BLIND-AUDITED

Benchmark numbers do not transfer to real data, and we say so with measurements. The 114 Hebrew statements of a production question were replayed in original arrival order. Production had one 59-member catch-all theme, one 24-member "synthesis" and 14 unplaced statements; the replay produced 11 precise syntheses (2–8 members), 2 themes and 27 statements deliberately left open, with the old 24-member blob reappearing as one proposal holding exactly its two true paraphrases. With no ground truth, we ran a verification battery: two blind adversarial merge auditors, two missed-merge sweepers (only cross-judge-agreed findings count) and a text-fidelity scorer. Member-level merge precision was 22/34 (65%), one 8-member synthesis had snowballed 6 wrong members, and 19 of 21

Table 2: Bare-LLM baselines vs the production pipeline on the 100-statement English corpus, identical arrival orders and scorer; mean (range) over seeds $\{42, 7, 1234\}$. Pipeline row from the certified runs of §5.2; its wall-clock is dominated by the harness's settle waits and is not comparable.

Condition	Composite	Synth F1	Pairs clean / false	Topic F1	Calls; tokens	Cost / run
L1 batch, worker model	0.933 (0.918–0.949)	1.000	50/50; 0	0.832 (0.796–0.873)	1; 3.8k	\$0.003
L1 batch, strong model	0.937 (0.918–0.955)	1.000	50/50; 0	0.842 (0.796–0.887)	1; 3.8k	\$0.028
L2 incremental, worker	0.965 (0.937–0.984)	1.000	50/50; 0	0.911 (0.844–0.960)	101; 100k	\$0.029
Production pipeline	0.900 (0.884–0.910)	1.000	50/50; 0	0.750 (0.711–0.775)	$\sim$2.85/stmt, $O(k)$ tokens	–

misses were silent. Three fixes followed (the anti-snowball transitivity check, the revisit pass, writer clauses forbidding added commitments) and the replay was repeated: precision 32/43 (74%), no synthesis above 4 members, grouped statements $34 \to 43$, fidelity $0.647 \to 0.953$, and the silent-miss class abolished. We do not quote the benchmark's 0.93 as the production expectation.

5.4 LLM-ONLY BASELINES ON THE LIVE CORPUS

To test the bare-LLM shortcut on our own instrument, we ran two conditions over the English 100-statement corpus with the same three arrival orders and scorer, and no embeddings, queue, judges or sweeps. **L1, one-shot batch**: one prompt with all 100 statements returns syntheses and topics (worker and strong model). **L2, incremental**: statements arrive one at a time; each call sees every current synthesis with all member texts verbatim and answers join/new with a confidence (worker model, 101 calls per run). Table 2 reports the result, and it is not the one a reader of §3.2 might expect.

The synthesis task is saturated by this corpus. Every condition, model and seed produced the identical synthesis partition, the 50 ground-truth pairs and nothing else (cross-seed ARI 1.000; all 300 of L2's join/new decisions at confidence ≥ 0.86), exactly as the pipeline does. *The bare model beats the pipeline on theming*, chiefly because it sees the whole set at once: the pipeline's residual is fragmentation (10–17 headings for a true 10), whereas the bare model never produced more than one heading per true theme, and its one systematic loss is a defensible taxonomy disagreement (street trees and community gardens under "environment" rather than "parks"). The strong model bought nothing at $10\times$ the price.

Three conclusions follow. First, **the pipeline's justification is operational, not per-decision accuracy**: on a clean corpus that fits in one prompt, judgement alone is exact, which is the "judgement disposes" half of our law. What geometry buys is the other half: an online form for the layer that must be order-stable; $O(k)$ rather than $O(n)$ tokens per arrival (L2's per-step prompt grew linearly to 1.6k tokens at n=100, so its total cost is $\Theta(n^2)$ and its prompts reach the context ceiling and the list-scale regime of Table 1 at 10^3–10^4 statements); robustness where hard pairs exist (§5.2, §5.3); and bounded, cached spend under bursts. Second, **a benchmark that separates the methods at the synthesis layer remains to be built**: well-formed English paraphrase pairs stress no judge, and the real-data replay, where precision is 74%, is where judges, gates and sweeps earn their keep. Third, **theming should be batch, synthesis should be online**: themes only organise the page and are not what participants agreed to, so rebuilding them from the $\leq$50 current syntheses in one cheap call is admissible where rebuilding syntheses is not; L2's topic pass over tidy syntheses (F1 0.91) is the design we intend to adopt in place of incremental filing.

Cost and latency. Embeddings are a rounding error next to completions. The registry classifier costs $\approx \$0.11$ per 1,000 classifications at small codebooks and $\approx \$0.41$ at $\sim$100-claim codebooks; the full engine in corpus replay averaged 2.85 calls per statement, $\approx$ \$0.2–0.5 per 1,000 statements, at 1–2 s per placement (a spawn with writer 8–10 s). The verdict cache and miss-billed budgets are not optional at scale: the sweep incident above cost $\approx$3,100 completions per day on questions nobody was touching, and after the fix the same 21 refusals per tick that spanned 46–52 s span $\sim$1 s. Per-arrival triggers under a burst remain unbounded and fail closed (a duplicate left visible, never a wrong merge).

6 RELATIONSHIP TO PREFERENCE GEOMETRY

Blair et al. and this work reach one diagnosis from opposite directions and adopt opposite remedies. They *repair the geometry*, learning an encoder or a per-topic projection whose cosine *is* approximately the utility margin, preserving $O(1)$ scoring, differentiability and compatibility with facility-location and fair-clustering machinery (Chen et al., 2019; Micha & Shah, 2020). We *demote* it: cosine retrieves and ranks, a discrete judgement decides, and the registry replaces "position in a metric space" with "membership in a named equivalence class" as the unit of meaning. On the shared instrument, judged placement reaches 95–98% where the best repaired geometry reaches 80–81%; the costs run the other way (1–2 calls per placement, no utility estimates for unvoted statements, the judge's biases). The tasks are siblings, not twins: Blair et al. model *preferential similarity* (would the same participant endorse both?), our merge relation is *propositional equivalence* (are these the same proposal?). They coincide on hard triplets, which licenses the shared benchmark, and diverge elsewhere: two distinct proposals can be preferentially similar (a voter who wants speed bumps likely also endorses lower limits) and we must *not* merge them, only theme them; equivalence is binary and symmetric and fits a discrete codebook, preference is graded and participant-relative and wants a geometry. One relation neither geometry can carry is *opposition*: "ban X" and "permit X" are near in any topical space, and even a preference space encodes disagreement as distance, not claim–counterclaim structure; we store it as a typed edge, which lets the interface show an idea together with its organised opposition.

Because our architecture confines geometry to candidate generation, their contribution slots in without redesign: a preference-tuned or per-topic-projected encoder is simply a better candidate generator. Cosine-ordered candidate lists are today *adversarially ordered* (distractors outrank matches in 863/880 cases), so a stance-faithful ranker should close the residual cross-lingual recall gap, shrink gray zones (cutting judge calls), and exploit the continuous evaluations we collect on every statement, which their per-topic projection turns into a preference geometry from $\sim$50 labelled triplets. What flows back is the decision layer: their downstream problems still need statements deduplicated, labelled and synthesised first, and judged, fail-closed, audit-logged placement over a proposed geometry is one concrete answer to their own warning that the embedding should not ground binding decisions. The registry's EXPRESSES/OPPOSES verdicts are, incidentally, the (anchor, match, distractor) triplets their tuning consumes, generated in-domain for free.

7 Related Work

Text-based collective decision-making aggregates over some geometry of participants or statements (Small et al., 2021; Fish et al., 2026; Chooi et al., 2026; De et al., 2026); Tessler et al. (2024) and Bakker et al. (2022) generate consensus statements with LLMs and preference data; Small et al. (2023) and Konya et al. (2023) survey LLM roles in deliberation. Sentence encoders are trained and benchmarked on semantic tasks (Reimers & Gurevych, 2019; Ni et al., 2022; Muennighoff et al., 2023); their blindness to negation (Vahtola et al., 2022) and stance- or contradiction-aware variants (Ghafouri et al., 2024; Xu et al., 2024; Gao et al., 2021) address the representation, as do Blair et al.. LLM-guided clustering (Viswanathan et al., 2024; Zhang et al., 2023; Pham et al., 2024; Wan et al., 2024) is batch over a fixed corpus; our setting is online. LLM-as-judge reliability, position bias and prompt sensitivity (Zheng et al., 2023; Wang et al., 2023; Liu et al., 2024; Sclar et al., 2024; Atil et al., 2024; Chen et al., 2023) motivate our short ranked lists, judge-once fingerprints, verdict versioning and upgrade regression tests. Retrieve-then-rerank (Nogueira & Cho, 2019; Lewis et al., 2020) is the nearest architectural relative; ours returns a typed relation, never a score, against a growing codebook rather than the corpus. Incremental clustering (Charikar et al., 2004) and record linkage (Fellegi & Sunter, 1969) anticipate the propose-then-decide split.

8 Limitations and Conclusion

Benchmark corpora are partly synthetic; Blair et al.'s natural-data numbers (65–69%) suggest most real pairs are easy and the hard tail is what the architecture is for, and the real-data replay shows precision of 74% where the benchmark shows 100%. Theming is the open front (§5.4 indicates the fix, not yet measured live). Judge, writer and canonicaliser share a vendor's model family; the 5% cross-model audit estimates drift, not family-correlated bias. Costs are not held to a fixed token budget, and evaluation is English and Hebrew only. Within those limits the conclusion is firm: a live deliberation platform cannot trust embedding distance to decide identity, and cannot afford, stabilise or audit a bare LLM as an online clustering algorithm at scale, even though on a small clean corpus that LLM is exact. The architecture that works lets geometry propose a short ranked candidate set, lets a discrete, fail-closed, cached judgement dispose with no exceptions, turns deduplication into classification against a maintained codebook, and rebuilds in batch only the layer that need not be order-stable. Its synthesis layer is exact and order-independent on our benchmarks and beats a purpose-tuned preference geometry by 15–18 points on that geometry's own hard instrument; the two approaches compose rather than compete.

AI use statement

Generative AI tools were used in this work for: writing and editing the manuscript text under the authors' direction; writing and refactoring parts of the production system and the evaluation harnesses (verified by tests, by importing production functions unmodified into harnesses, and by build fingerprints recorded per run); generating the synthetic benchmark corpora (a second corpus was generated by a different vendor's model specifically to test cross-generator generalisation); and, as the object of study, the judge, writer, canonicaliser and theme-assignment components themselves. Generative AI was also used as blind auditors in the real-data verification battery (§5.3), where only findings agreed by independent judges were counted. We have reviewed all AI-assisted work, checked every reported number against the committed run artefacts, and take responsibility for the final content.

Ethics statement

The system operates on participant statements in public deliberations; it never rewrites a participant's own text (synthesised texts are new statements that replace pooled duplicates, and a fidelity audit checks that each member's intent remains visible). The real-data replay used a read-only export of one production question with no identifying information; no statements are quoted. A wrong merge misrepresents a participant, which is why the architecture's doubt biases, fail-closed defaults, review queue and typed opposition edges exist; we consider the false-merge rate, not accuracy, the primary safety metric and report it throughout.

Reproducibility statement

The pipeline source, the hard-triplet harness (which imports production functions unmodified), the 100-statement corpora in both languages, the scorer, every run's inputs and outputs, the blind-audit artefacts, and the LLM-only baseline scripts (9 runs, \$0.18 total) are committed in the project repository (to be linked in the camera-ready). Appendix A lists all thresholds; Appendix B gives the judge rubrics; Appendix F documents the measurement discipline (build certification, settle windows, contaminated runs marked do-not-cite) without which two of our own conclusions had inverted.

References

Berk Atil, Alexa Chittams, Liseng Fu, Ferhan Ture, Lixinyu Xu, and Breck Xu. LLM stability: A detailed analysis with some surprises. *arXiv preprint arXiv:2408.04667*, 2024.

Michiel A. Bakker, Martin J. Chadwick, Hannah Sheahan, Michael Henry Tessler, Lucy Campbell-Gillingham, Jan Balaguer, Nat McAleese, Amelia Glaese, John Aslanides, Matthew Botvinick, and Christopher Summerfield. Fine-tuning language models to find agreement among humans with diverse preferences. In *Advances in Neural Information Processing Systems*, volume 35, pp. 38176–38189, 2022.

Carter Blair, Ariel D. Procaccia, and Milind Tambe. Embeddings for preferences, not semantics. *arXiv preprint arXiv:2605.08360*, 2026.

Niclas Boehmer, Sara Fish, and Ariel D. Procaccia. Generative social choice: The next generation. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 4627–4659, 2025.

Moses Charikar, Chandra Chekuri, Tomás Feder, and Rajeev Motwani. Incremental clustering and dynamic information retrieval. *SIAM Journal on Computing*, 33(6):1417–1440, 2004.

Lingjiao Chen, Matei Zaharia, and James Zou. How is ChatGPT's behavior changing over time? *arXiv preprint arXiv:2307.09009*, 2023.

Xingyu Chen, Brandon Fain, Liang Lyu, and Kamesh Munagala. Proportionally fair clustering. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pp. 1032–1041, 2019.

Jay Chooi, Paul Gölz, Ariel D. Procaccia, Benjamin Schiffer, and Shirley Zhang. Finding common ground in a sea of alternatives. *arXiv preprint arXiv:2603.16751*, 2026.

Soham De, Lodewijk Gelauff, Ashish Goel, Smitha Milli, Ariel D. Procaccia, and Alice Siu. Question the questions: Auditing representation in online deliberative processes. In *Proceedings of the ACM Web Conference 2026*, pp. 1640–1650, 2026. doi: 10.1145/3774904.3792474.

Ivan P. Fellegi and Alan B. Sunter. A theory for record linkage. *Journal of the American Statistical Association*, 64(328):1183–1210, 1969.

Sara Fish, Paul Gölz, David C. Parkes, Ariel D. Procaccia, Gili Rusak, Itai Shapira, and Manuel Wüthrich. Generative social choice. *Journal of the ACM*, 73(2), 2026. doi: 10.1145/3799709.

Tianyu Gao, Xingcheng Yao, and Danqi Chen. SimCSE: Simple contrastive learning of sentence embeddings. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6894–6910, 2021. doi: 10.18653/v1/2021.emnlp-main.552.

Vahid Ghafouri, Jose Such, and Guillermo Suarez-Tangil. I love pineapple on pizza $\neq$ i hate pineapple on pizza: Stance-aware sentence transformers for opinion mining. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 21046–21058, 2024. doi: 10.18653/v1/2024.emnlp-main.1171.

Andrew Konya, Lisa Schirch, Colin Irwin, and Aviv Ovadya. Democratic policy development using collective dialogues and AI. *arXiv preprint arXiv:2311.02242*, 2023.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pp. 9459–9474, 2020.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. doi: 10.1162/tacl_a_00638.

Evi Micha and Nisarg Shah. Proportionally fair clustering revisited. In *47th International Colloquium on Automata, Languages, and Programming (ICALP)*, volume 168 of *LIPIcs*, pp. 85:1–85:16, 2020.

Niklas Muennighoff, Nouamane Tazi, Loic Magne, and Nils Reimers. MTEB: Massive text embedding benchmark. In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pp. 2014–2037, 2023. doi: 10.18653/v1/2023.eacl-main.148.

Jianmo Ni, Gustavo Hernandez Abrego, Noah Constant, Ji Ma, Keith B. Hall, Daniel Cer, and Yinfei Yang. Sentence-T5: Scalable sentence encoders from pre-trained text-to-text models. In *Findings of the Association for Computational Linguistics: ACL 2022*, pp. 1864–1874, 2022. doi: 10.18653/v1/2022.findings-acl.146.

Rodrigo Nogueira and Kyunghyun Cho. Passage re-ranking with BERT. *arXiv preprint arXiv:1901.04085*, 2019.

OpenAI. New embedding models and API updates. https://openai.com/index/new-embedding-models-and-api-updates/, 2024.

Chau Minh Pham, Alexander Hoyle, Simeng Sun, Philip Resnik, and Mohit Iyyer. TopicGPT: A prompt-based topic modeling framework. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pp. 2956–2984, 2024.

Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, pp. 3982–3992, 2019. doi: 10.18653/v1/D19-1410.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024.

Christopher T. Small, Michael Bjorkegren, Timo Erkkilä, Lynette Shaw, and Colin Megill. Polis: Scaling deliberation by mapping high dimensional opinion spaces. *RECERCA. Revista de Pensament i Anàlisi*, 26(2):1–26, 2021. doi: 10.6035/recerca.5516.

Christopher T. Small, Ivan Vendrov, Esin Durmus, Hadjar Homaei, Elizabeth Barry, Julien Cornebise, Ted Suzman, Deep Ganguli, and Colin Megill. Opportunities and risks of LLMs for scalable deliberation with Polis. *arXiv preprint arXiv:2306.11932*, 2023.

Michael Henry Tessler, Michiel A. Bakker, Daniel Jarrett, Hannah Sheahan, Martin J. Chadwick, Raphael Koster, Georgina Evans, Lucy Campbell-Gillingham, Tantum Collins, David C. Parkes, Matthew Botvinick, and Christopher Summerfield. AI can help humans find common ground in democratic deliberation. *Science*, 386(6719):eadq2852, 2024. doi: 10.1126/science.adq2852.

Teemu Vahtola, Mathias Creutz, and Jörg Tiedemann. It is not easy to detect paraphrases: Analysing semantic similarity with antonyms and negation using the new SemAntoNeg benchmark. In *Proceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*, pp. 249–262, 2022. doi: 10.18653/v1/2022.blackboxnlp-1.20.

Vijay Viswanathan, Kiril Gashteovski, Carolin Lawrence, Tongshuang Wu, and Graham Neubig. Large language models enable few-shot clustering. *Transactions of the Association for Computational Linguistics*, 12:321–333, 2024.

Mengting Wan, Tara Safavi, Sujay Kumar Jauhar, Yujin Kim, Scott Counts, Jennifer Neville, Siddharth Suri, Chirag Shah, Ryen W. White, Longqi Yang, Reid Andersen, Georg Buscher, Dhruv Joshi, and Nagu Rangan. TnT-LLM: Text mining at scale with large language models. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, pp. 5836–5847, 2024.

Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. Large language models are not fair evaluators. *arXiv preprint arXiv:2305.17926*, 2023.

Haike Xu, Zongyu Lin, Yizhou Sun, Kai-Wei Chang, and Piotr Indyk. SparseCL: Sparse contrastive learning for contradiction retrieval. *arXiv preprint arXiv:2406.10746*, 2024.

Yuwei Zhang, Zihan Wang, and Jingbo Shang. ClusterLLM: Large language models as a guide for text clustering. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 13903–13920, 2023.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems 36, Datasets and Benchmarks Track*, 2023.

A Bands, constants and failure policies

Other constants: $k = 15$ neighbours; 3 spawn candidates tried per arrival; consolidation every 15 placements; confirmed at ≥ 3 members; 2 unchecked broadenings before the anchor check; registry confidence floor 0.6; 5% shadow audit; sweep budget 25 judge calls per question per tick, billed on cache misses; hierarchical classification above 30 claims with at most 2 routed topics; 20 pairs per equivalence call.

Table 3: Cosine bands (defaults for `text-embedding-3-small`; a separately calibrated set ships for `-3-large`, because bands are calibrated to a geometry, not a pipeline). Crossing a band makes a statement a candidate for an action, never triggers it.

Band	3-small	3-large	Role
attachThreshold	0.85	0.86	candidate may be a paraphrase of an existing synthesis
synthLowerBound	0.78	0.80	pair may warrant a new synthesis
clusterThreshold	0.60	0.75	topical relatedness (theming)
reviewLowerBound	0.45	–	ANN floor; below this nothing is a candidate
synth centroid floor	0.815	–	halfway up the synthesis band (§4.2)
re-judge proposal	0.82	–	cross-member top-2 evidence; deliberately below decision grade

Table 4: Failure policy by decision type. The invariant: an infrastructure failure may delay cleanup but must never scatter an equivalence class, and must never masquerade as a burst of honest negative verdicts.

Decision	On judge error / timeout	Why
Merge / attach / spawn	fail closed (*different*; nothing merges)	a wrong merge is permanent for every reader
Re-validation of an existing cluster	fail open (members kept)	scattering a class is worse than delaying cleanup
Registry classification	*none* + `failedClosed` marker	outages must be separable from "this is new"
Claim mutation classification	fail closed to *different*	forces batched member re-validation
Verdict cache	fallback verdicts never written	a transient failure must not poison future runs
Debounced or failed spawn	re-enqueued, never dropped	a suppressed action must have a custodian

B JUDGE RUBRICS (ABRIDGED FROM THE SHIPPED PROMPTS)

Equivalence judge. *same*: A and B are paraphrases of essentially the same proposal; their authors would probably agree they meant the same thing. *related*: same topic but different actions, stances or magnitudes. *different*: about different things; wording similar by coincidence. *opposite*: contradictory actions on the same subject ("ban X" vs "permit X"). Use *same* ONLY when the two proposals are essentially interchangeable. When in doubt between *same* and *related*, prefer *related*. Verdicts are returned as JSON for up to 20 pairs; pairs omitted from the response receive *different*.

Registry classifier. Statements may be in any language; judge meaning, not wording. Each claim line gives the canonical claim, a plain-language explanation and an example member statement; use them to judge the claim's precise meaning. CAUTION: a statement may reuse an example's exact wording while taking the OPPOSITE stance (an inserted "not", a swapped verb). Shared phrasing is never evidence of a match. *expresses*: the statement proposes essentially the same thing as one claim; its author would agree the claim states their idea. *opposes*: contradicts a claim; report it but it is not a match. *none*: nothing listed covers it, or it only shares a topic while proposing a different action, stance or magnitude. When in doubt between *expresses* and *none*, answer *none*. Output: match index, relation, confidence, brief reason.

Prompt sensitivity, measured. On 14 triplets with the architecture held constant, a plausible hand-written classifier prompt scored 36% triplet accuracy, 43% false merges and 36% counter-edges; the production prompt scored 79%, 7% and 79%. The difference is the stance caution, the doubt rule and the confidence floor. An evaluation harness must import the production classifier, never approximate it.

Synthesis writer (refusal stage). Refuse if the inputs pull opposite ways on the same lever (directional conflict), or if they are distinct specific interventions that happen to address related problems. Accept only true paraphrases of the same specific action: same lever, same direction, same intervention; if a supporter could reasonably back one while opposing the other, the inputs are distinct ideas. Formulation differences (category vs instance, level of detail) are not grounds for refusal; refuse only when the difference IS the intervention. Never widen scope beyond what the inputs say; never add commitments the inputs do not make; every input must remain visible in the output. When in doubt, refuse.

Theme judge. You see each existing theme with its contents. File the proposal under the theme whose contents it belongs with, or answer NONE. When unsure, answer NONE: a topic created too eagerly is cheap (the consolidation sweep merges duplicate headings), a proposal filed under the wrong topic stays there for every reader. You may not create a theme.

Claim canonicaliser. The claim MUST carry the author's stance: direction (for/against, allow/ban, expand/reduce) and any condition that defines it. Never restate a position from its exception's point of view: "X should be illegal except in case C" must not become "X is permitted in case C". Acceptance test: the author would say "yes, that is my position" and someone holding the opposite position would reject it. Write in the language of the question.

C CONVERGENCE LAYER: DETAILS

The queue worker runs every minute and drains deferred work, including spawns deferred by a per-*pair* debounce (originally per-question, which suppressed spawns for unrelated pairs and silently dropped them; 45 attempts = 24 spawned + 21 dropped, capping pair recall at 40% although every pair was inside the attach band) and spawns that failed after the gate (an LLM error, a malformed proposal, a transient write failure), which previously ended the pipeline with no audit row and no retry. The re-judge sweep runs every 10 minutes over at most a bounded number of questions, proposes cross-synthesis merges wherever cross-member top-2 evidence exceeds 0.82, memoises rejected pairs per sweep, confirms each proposal with the cached equivalence judge, requires the farthest cross-member pair to be judged SAME (the anti-snowball transitivity check, up to 6 members per side), and only then consults the writer; it is bounded by 10 merges and 25 judge calls (cache misses) per question per tick. A revisit pass re-enqueues left-behind statements whose evidence clears the synthesis band, keyed on a fingerprint of the question's cluster state so a settled question is not re-ground. Theme consolidation merges duplicate headings judged on their members with at most 8 merges per sweep and at least 3 themes present; each distinct theme set is judged once, keyed on a fingerprint of which headings exist and how much each holds. Claim consolidation (registry) runs every 15 placements as described in §4.5. A separate bulk path exists for admin-initiated full re-synthesis (graph components over 0.92-cosine edges, medoid-based two-tier judging with auto-accept above 0.94 and auto-reject below 0.60, an LLM-call cap of $\min(2000, 0.2N)$) and is kept behind feature flags; a controlled validation of that path on five corpora with known structure recovered ground-truth syntheses exactly on realistically-worded input and surfaced three production defects, and its dominant residual failure was corpus realism (maximal lexical variation fragmenting recovery).

D ROBUSTNESS ACROSS MODEL GENERATIONS: FULL PILOT-150 REPLICATION

Because the numbers above were produced on one model generation, we re-ran the registry conditions on a fixed, stratified 150-triplet subsample after a model upgrade, canonicals regenerated by the new model, contrasts paired on identical triplets. Raw-anchor classification is robust: 96.7% [92.4, 98.6] vs 98.7% before ($p = 0.45$). The enriched-canonical condition regressed significantly, 92.0% $\rightarrow$ 79.7% (21 vs 3 discordant, $p = 2.8 \times 10^{-4}$), to parity with DPT, and every inspected false attach decomposed into the canonicalisation link: the new model's canonical had *dropped the anchor's stance qualifier* ("illegal, with a life exception" became "permitted when the mother's life is at risk"), after which the classifier *correctly* matched the stance-flipped distractor to a stance-lossy claim. A reasoning-token headroom fix (on 2/150 long inputs the model deterministically exhausted the completion cap before emitting JSON, and fail-closed held) took raw-anchor to 98.0% with no regressions; a stance-preserving generation prompt recovered only part of the enriched loss (82.7%), because a multi-part opinion compresses to one claim and the flipped rewrite may genuinely express the surviving point. Compressive canonicalisation is inherently lossy; the robust configuration keeps generated canonicals for *display*, weights the verbatim exemplar as the operative matching text, and regression-tests canonicalisation on every model upgrade.

E THE LIVE-BENCHMARK CAMPAIGN ARC

F DOCUMENTED FAILURE MODES AND MEASUREMENT DISCIPLINE

The topic-cluster black hole. In the first instrumented run one early topical cluster absorbed all 100 statements and no synthesis formed. The mechanism is general: in a corpus whose cosine floor is

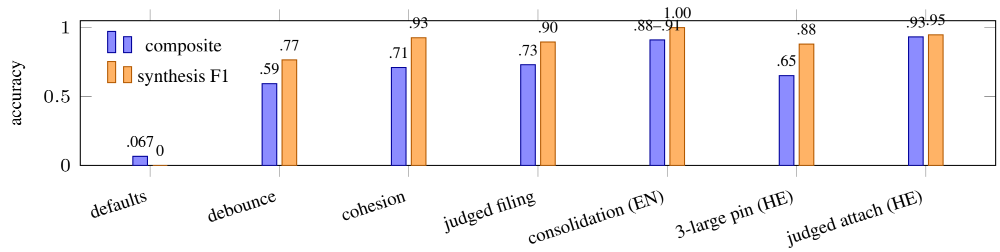

Figure 2: The live-benchmark arc, cumulative configurations at shipped thresholds (English then Hebrew). The English headline is quoted as a range over three arrival orders.

lifted (here by the shared question prefix), the first cluster created above the topical threshold becomes an attractor every subsequent statement clears, and if membership is terminal every paraphrase pair is foreclosed. Coverage was 100% while pair precision was 0.091. The fix was structure, not a threshold: conjunctive cohesion gates, non-terminal topic membership, judged theme filing, and disabling raw-pair theme creation. The black hole reappeared, instructively, when the Hebrew embedding model was upgraded without recalibrating the bands (a 45-member cross-topic theme, 25 members entering through the one judge-free attach path), confirming that thresholds are properties of a geometry.

Snowballing. Attach-on-best-single-member lets a synthesis grow by a chain of pairwise links until it spans several ideas. Measured in production and again on the real-data replay (an 8-member synthesis with 6 wrong members). Answered by the top-2 member evidence, the centroid$\wedge$quorum cohesion gate, and the farthest-pair transitivity check in the sweep.

Silent drops in the streaming path. A 15-second spawn debounce keyed per question suppressed spawns for unrelated pairs (45 attempts = 24 spawned + 21 dropped) and capped pair recall at 40% even though every pair was inside the attach band; a spawn that failed after the gate left no audit row and no retry. Both are instances of one shape, a suppressed action with no custodian, and both fixes are re-enqueues.

Order of passes. Theming before synthesis let the general claim win on the specific claim's evidence (17 of 23 topical attachments justified by the statement's own twin). Most-specific-first fixed it.

Looping a non-deterministic judge. A scheduled sweep asked to *find* merge groups in a settled, already-correct set of ten headings proposed a wrong merge in one of two samples; production would have asked $\sim$144 times a day. Each distinct set is now judged once, keyed on a fingerprint.

Replaying on end states predicts nothing. An offline replay of the consolidation judge on each run's *final* theme set predicted a clean 11 $\rightarrow$ 10; live, the sweep faced 19 half-formed headings and merged 9 in one pass, and the run scored worse. A subsequent reconstruction from full before/after membership snapshots showed the impurity had entered at *filing*, not consolidation, in every run, so the original revert was decided on confounded evidence. Decisions must be evaluated on the states they actually encounter.

Measurement that fails toward plausibility. A harness that *re-implemented* the production merge sweep kept measuring the copy after the production code gained an LLM gate; a settle detector's quiet window could not distinguish "queued" from "finished" and reported 47/50 for a run that achieved 50/50. Neither produced an error; both produced a number. The campaign's response is a reproducibility discipline we commend to anyone benchmarking live pipelines: import production functions unmodified, certify builds byte-for-byte before and after each run, require two consecutive quiet windows and an unchanged double export, and mark contaminated runs "do not cite" rather than exclude them.

G THEME FILING: THE 2×2 AND THE DECISION RULE

On 58 judged filing decisions reconstructed to their exact live contexts (3 repeats each, the live model), variants were: F0 titles + prefer-file (ships at the time): 66.1% accuracy, 28 misfiles, 31 over-NONEs; F1 contents + prefer-file: 64.4%, 37, 25; F2 contents + unsure→NONE: 62.6%, 17, 48; F3 titles + unsure→NONE: 39.7%, 15, 90; F4 contents + conflict→NONE: 70.1%, 30, 22; F5 contents + NONE with narrowness exception: 71.8%, 23, 26. The registered rule (minimise 3·misfiles + over-NONEs, hard cap misfiles ≤ 19) selects F2; F4/F5 raise accuracy by re-licensing misfiles and fail the cap. Live confirmation: misfiles $12 \rightarrow 5$ ($20.7\% \rightarrow 9.3\%$) with per-decision accuracy flat, and the reversible errors were then actually repaired by the sweep (13 donor headings merged, 12 same-topic).

H TEXT FIDELITY

A synthesis replaces what two people wrote with wording a model produced, and every membership metric would read 1.000 for a merge that held the right members and published a proposal carrying only one of them. A validated fidelity judge (self-tested on five hand-written merges with known answers) grades each member as preserved / weakened / lost, plus scope inflation and added commitments. Title-only scoring: 0.84–0.92 fidelity, zero members lost across 300 member statements in three runs. Full-text scoring over the real compiled writer: fidelity 0.990 with 4/50 scope inflations, caused by a contradiction in the prompt (forbid inventing facts, but state who does what on what timeline); making that instruction conditional on the inputs and giving scope its own prohibition took inflation to 0/50 and fidelity to 1.000 with no new refusals. On the real-data replay, fidelity moved $0.647 \rightarrow 0.953$ after the writer clauses.